

1 Worksheet

Question 1.1 (Practice 1). Let $f \in L^2$ with $\sum_n |\hat{f}(n)| < \infty$. Show that there is a continuous function g such that $f = g$ a.e.

Proof. Let g be f 's Fourier series. This gives $\|f - g\|_{L^2} = 0$, so $f = g$ a.e., and it remains to show that g is continuous. Since $\sum_n |\hat{f}(n)| < \infty$, g converges absolutely and uniformly, so g is indeed continuous. $\square$

Question 1.2 (Practice 2). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be 2π -periodic and in C^α . That is, $|f(x) - f(y)| \leq C|x - y|^\alpha$ for some $C > 0$ and $\alpha \in (0, 1)$. Show that the Fourier series of f converges pointwise to f .

Proof. $\square$

Question 1.3 (Practice 3). Show that L^∞ is complete.

Proof. Let f_n be a Cauchy sequence in L^∞ . We can pass to a subsequence such that $\|f_k - f_{k+1}\|_\infty \leq 2^{-k}$ via a fast Cauchy argument. Let $f(x) = f_1(x) + \sum_{k=1}^\infty (f_{k+1}(x) - f_k(x))$. The series converges absolutely a.e. since each term is bounded by 2^{-k} a.e., so f exists a.e., and it shows that f is L^∞ . For convergence, notice that

$$\|f_n - f\|_\infty = \left\| f_{n+1}(x) + \sum_{k=n+1}^\infty (f_{k+1}(x) - f_k(x)) \right\|_\infty$$

which approaches zero by our bound. $\square$

Question 1.4 (Practice 4). Let $f : E \rightarrow \mathbb{R}$. If either $m(E) < \infty$ or $f \in L^p(E)$ for some $p \geq 1$, show that

$$\lim_{p \rightarrow \infty} \|f\|_{L^p(E)} = \|f\|_{L^\infty(E)}.$$

Find a counterexample if the assumptions are dropped.

Proof. $\square$

Question 1.5 (Practice 5). Let $p \in [1, \infty)$. Let A_p be the set of functions $f : [0, 1] \rightarrow \mathbb{R}$ that have a continuous derivative with

$$\int_0^1 |f'(x)|^p dx \leq 1.$$

For which p are closed (with respect to the topology of uniform convergence) subsets of A_p compact?

Proof. $\square$

Question 1.6 (Practice 6). Let $p \in [1, \infty]$. Construct a function $f \in L^p(\mathbb{R})$ but $f \notin L^q(\mathbb{R})$ for any $q \neq p$.

Proof. $\square$

Question 1.7 (Practice 7). Let f_n be a sequence that converges in L^p to some f . Show that there exists a subsequence that converges a.e.

Proof. $\square$

Question 1.8 (Practice 8). This problem explores the concept of *convergence in measure*. Let $f_n : E \rightarrow \mathbb{R}$ be a sequence of measurable functions. We say that f_n converges to some f in measure if for all $\varepsilon > 0$ we have

$$\lim_{n \rightarrow \infty} m(\{x : |f_n(x) - f(x)| > \varepsilon\}) = 0.$$

- (a) If f_n converges to f in L^p show that it converges in measure.
- (b) If a sequence converges in measure, show that a subsequence converges pointwise a.e.
- (c) If $m(E) < \infty$ and f_n converges pointwise, show that it converges in measure.
- (d) Find sequences that converge in measure but not pointwise.
- (e) Show that Fatou's Lemma and the MCT hold if pointwise convergence is replaced by convergence in measure.

Proof.

□

Question 1.9 (Practice 9). This problem shows that the Riemann integral coincides with the Lebesgue integral. In this problem let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function (not necessarily measurable).

- (a) Let P_k be a sequence of partitions of $[a, b]$ whose mesh goes to 0 and are nested inside each other (i.e., P_{k+1} is a refinement of P_k for all k). For P_k let

$$G_k(x) = \sum_j M_j \mathbf{1}_{(t_{j-1}, t_j]}, \quad g_k(x) = \sum_j m_j \mathbf{1}_{(t_{j-1}, t_j]},$$

where the intervals $(t_{j-1}, t_j]$ are the partition of $[a, b]$ corresponding to P_k and M_j, m_j are the sup and inf of f over $[t_{j-1}, t_j]$. Show that the Lebesgue integrals of g_k and G_k correspond to the upper and lower Riemann sums associated to P_k and f .

- (b) Let $G(x) = \lim_{k \rightarrow \infty} G_k(x)$ and $g(x) = \lim_{k \rightarrow \infty} g_k(x)$ (why do the limits exist?). Note $g \leq f \leq G$ everywhere. Show that if f is Riemann integrable, then $g = G$ a.e. (simply note that the Lebesgue integrals of G and g are equal to the limits of the Lebesgue integrals of G_k and g_k which coincide if f is Riemann integrable, as these integrals equal the upper and lower Riemann sums associated to the partitions P_k). Therefore, f is measurable as g and G are limits of measurable functions and its Lebesgue integral equals its Riemann integral.

- (c) Define

$$H(x) = \lim_{\delta \rightarrow 0} \sup_{|y-x| \leq \delta} f(y), \quad h(x) = \lim_{\delta \rightarrow 0} \inf_{|y-x| \leq \delta} f(y).$$

Why do the above limits exist? Show that $H(x) = h(x)$ iff f is continuous at x .

- (d) Show that the Lebesgue integrals of H and h are the limits of the upper and lower Riemann sums of f over any sequence of partitions whose mesh goes to 0. So if f is Riemann integrable, $H = h$ a.e. Therefore f is the a.e. limit of continuous functions and hence is Lebesgue measurable and its Lebesgue integral equals its Riemann integral.

Proof.

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